

EXPERIMENT – 37

Implementation of Letter Frequency Attack on Monoalphabetic Substitution Cipher

CODE:

```
#include <stdio.h>

#include <ctype.h>

#include <string.h>


int main()
{
    char cipher[500], plain[500];
    int freq[26] = {0}, order[26], n, i, j, temp;

    char english[] = "ETAOINSHRDLCLUMWFGYPBVKJXQZ";

    printf("Enter ciphertext: ");
    fgets(cipher, sizeof(cipher), stdin);

    printf("Enter number of possible plaintexts: ");
    scanf("%d", &n);

    /* Count letter frequency */
    for (i = 0; cipher[i]; i++)
        if (isalpha(cipher[i]))
            freq[toupper(cipher[i]) - 'A']++;

    /* Arrange letters by frequency */
    for (i = 0; i < 26; i++)
```

```

    order[i] = i;

for (i = 0; i < 25; i++)
    for (j = i + 1; j < 26; j++)
        if (freq[order[i]] < freq[order[j]])
        {
            temp = order[i];
            order[i] = order[j];
            order[j] = temp;
        }

printf("\nPossible plaintexts:\n");

/* Produce candidates using frequency mapping */
for (int k = 0; k < n; k++)
{
    for (i = 0; cipher[i]; i++)
    {
        if (isalpha(cipher[i]))
        {
            int x = toupper(cipher[i]) - 'A';
            int pos = 0;

            for (j = 0; j < 26; j++)
                if (order[j] == x)
                    pos = j;

            plain[i] = english[(pos + k) % 26];
        }
    }
}

```

```

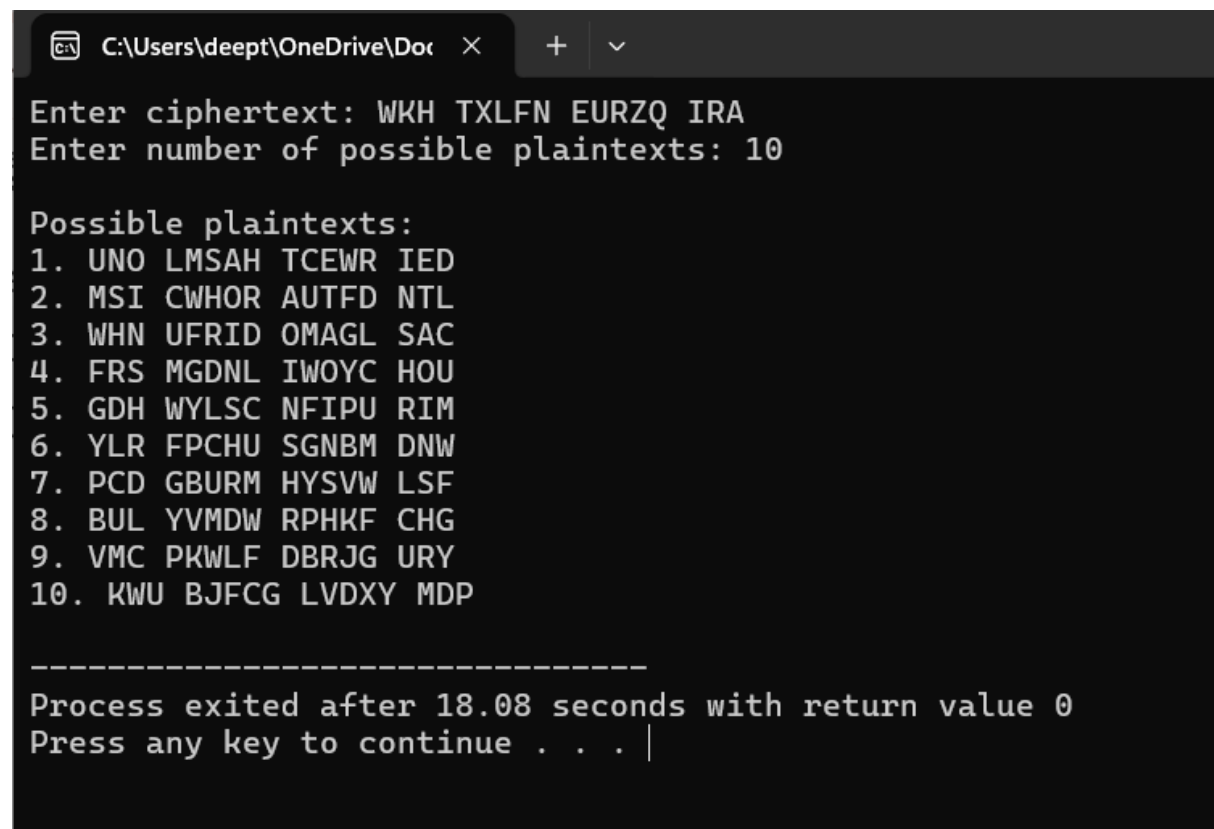
        else
            plain[i] = cipher[i];
        }

    plain[i] = '\0';
    printf("%d. %s", k + 1, plain);
}

return 0;
}

```

OUTPUT:



```

C:\Users\deept\OneDrive\Doc >
Enter ciphertext: WKH TXLFN EURZQ IRA
Enter number of possible plaintexts: 10

Possible plaintexts:
1. UNO LMSAH TCEWR IED
2. MSI CWHOR AUTFD NTL
3. WHN UFRID OMAGL SAC
4. FRS MGDNL IWOYC HOU
5. GDH WYLSC NFIPU RIM
6. YLR FPCHU SGNBM DNW
7. PCD GBURM HYSVW LSF
8. BUL YVMDW RPHKF CHG
9. VMC PKWLF DBRJG URY
10. KWU BJFCG LVDXY MDP

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Process exited after 18.08 seconds with return value 0
Press any key to continue . . .

```